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Reconstruction and flood simulation using GIS and Google Earth to determine the extent and damage of the January 14-15th 2018 ice jams on the Mohawk River in Schenectady, New York

Lauren Mahoney, Sally Louise Roscoe, Antonios Marsellos

Department of Geology, Environment, Sustainability at Hofstra University, Hempstead, NY

Introduction

During the 14th to the 15th of January 2018, the city of Schenectady located in Upstate New York experienced flooding due to the ice jam of the Mohawk River. The fluctuations of temperature in the month of January, ranging from extreme cold to then a period of warming, caused the ice jam to occur in the Schenectady region of the Mohawk River. As the floes blocked the flow of the river, the water became displaced. This resulted in flooding of the surrounding area. The damages were relatively minor. The flooding that occurred in the Stockade area of Schenectady in mid-January of 2018 was quite well documented through both local media and online newspapers. Previous work has been around spatial and time series prediction or damage evaluation at this location (Marsellos et al., 2010, Foster et al., 2011; Tsakiri et al., 2013, Lewis et al., 2016). This research utilizes Geographical Information System (GIS) software with Light Detection and Ranging (LiDAR), publicly announced locations showing the flood damage, and the related water level at the time of the flood (Fig. 1). A methodology is provided in this research on how to create a flood simulation to further understand the impact of the flood, as well as to prepare for the potential of the flood re-occurring within the given area.



Figure 1: Aerial image obtained from Google Earth of the study area and the utilized referenced points of flood occurrences with water level seen in publicly available pictures from the ice jam of the 14-15th of January 2018 in Schenectady, New York.

Methodology

Photographic evidence of the flooding was obtained from publicly available resources such as internet, online newspaper, blog-media, and we infer a direct measurement of the water level surface from the ice jam that occurred in the Schenectady region of the Mohawk River. The geographic locations of each of the photographs and the associated flood level were determined using Google Earth and GIS software. These research points were converted into .kmz files, which were then inserted into Global Mapper (GM), which is a GIS software. A LiDAR digital elevation model (LiDAR-DEM) was used as a virtual background for the research points, and once both the points and the LiDAR data was in GM, a 3D model was created. Two LiDAR-DEMs were constructed; one shows the Bare-Earth model (a surface of the Earth with no

objects on) and a DSM model (digital surface model; a surface of the Earth with all objects on). Flood simulations were run in both LiDAR-DEMs to compare the flood impacts.

The 3D model on Global Mapper (GM) was flooded multiple times in subsets of the studied area, using manually simulated water levels. These water levels were then determined by combining the visuals provided by the 3D model and the photographic evidence of the flood (Fig. 2 and Fig. 3). A table provided the measured water levels of the flood by the two techniques, that is the Google Earth and the LiDAR-DEM visualization is provided (Table 1).

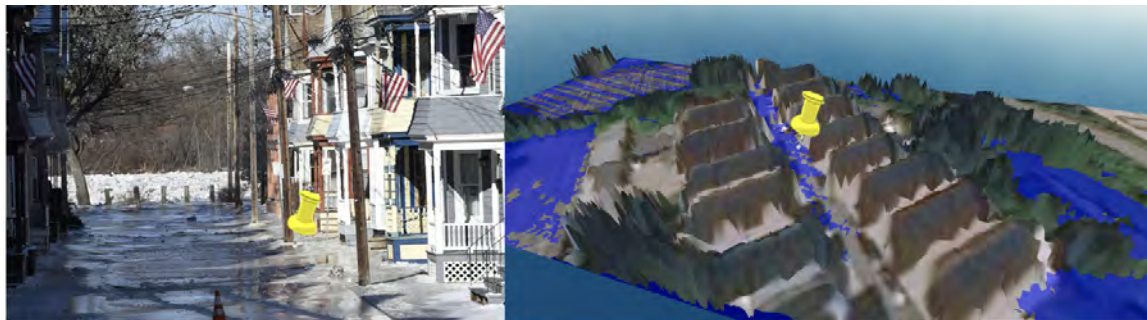


Figure 2: (left picture) Global Mapper 3D model of flooded Site 4, at Ingersoll Ave, and (right picture) the publicly available image showing the flood caused by the ice jam of January 2018 (images showing the flood were provided by Peter R. Barber for the Daily Gazette newspaper).

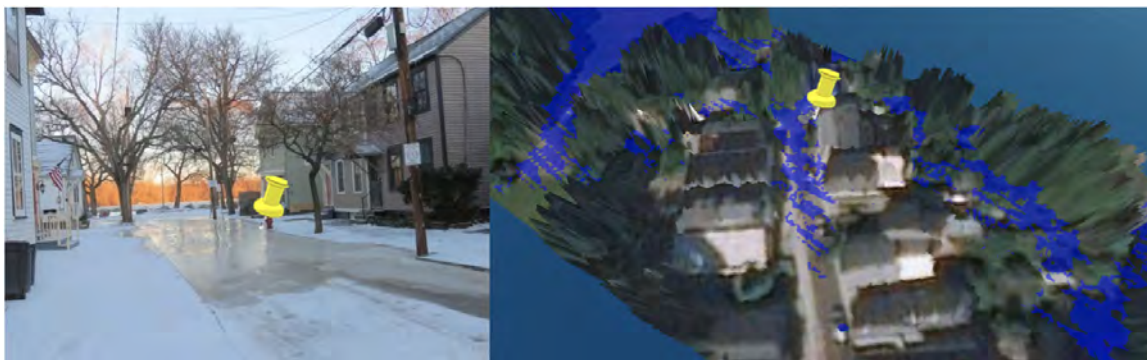


Figure 3: (left picture) Global Mapper 3D model of flooded Site 6, at North Street, and (right picture) the publicly available image showing the flood caused by the ice jam of January 2018 (images showing the flood were provided by David Giacalone for the Suns along the Mohawk blog).

Results

The Stockade neighborhood flooded fairly equally, with areas at lower elevations closer to the river, such as sites 1, 2, and 9, experiencing increased flooding. LiDAR-DEM using all LiDAR points were used to visualize flooding in 3D mode. The LiDAR-DEM using only the ground points (Bare Earth Model) was found to be inadequate to realistically visualize flood in the infrastructure of the town. The standard deviation of the measured water surface elevation from the flood images and the Global Mapper visualization is approximately 1.1 ft. while the average offset between those two techniques is 1.8 ft.

Table 1: Results and coordinates for each investigated site for determining the flood level using the LiDAR-DEM and the Google Earth elevation data.

Site	Address	Latitude	Longitude	Google Earth Elevation (feet, meters)		LiDAR (feet, meters)	
1	<i>Jumping Jacks Dine In</i>	42°49'19.77"	-73°57'23.53"	221.0	67.4	224.6	68.5
2	<i>Ferry St Water Gauge</i>	42°49'9.87"	-73°56'49.44"	221.0	67.4	223.4	68.1
3	<i>Rivers Casino</i>	42°49'27.89"	-73°56'9.09"	221.0	67.4	223.8	68.2
4	<i>Ingersoll Ave: Gray and White House</i>	42°49'13.00"	-73°56'38.61"	222.0	67.7	222.8	67.9
5	<i>Riverside Park</i>	42°49'16.92"	-73°56'49.70"	221.0	67.4	221.8	67.6
6	<i>North Street: Brown House</i>	42°49'10.83	-73°56'43.34"	223.0	68.0	223.6	68.2
7	<i>32 Ingersoll Ave</i>	42°49'13.99"	-73°56'40.16"	221.0	67.4	223.6	68.2
8	<i>Riverside Park Esplanade</i>	42°49'12.34"	-73°56'45.62"	221.0	67.4	223.0	68.0
9	<i>Schonowee Avenue</i>	42°49'17.24"	-73°57'5.92"	222.0	67.7	222.7	67.9

Discussion

The comparison of LiDAR data and Google Earth Pro data in this study displayed the contrast of these two different elevation datasets that those programs utilize for the flood simulation. The LiDAR data were more accurate for determining the elevations of our study areas, as well as for the simulation of the actual flood, which occurred in January 2018. However, Google Earth is more efficient in terms of fewer requirements for pre-processing of the elevation data and it is more widely accessible to the public. In addition, geolocation address was more feasible of our sites based on the photographs collected and the existing Google geodatabase.

The Bare Earth Model or Digital Terrain Model (DTM), is not an ideal model to use since it did not accurately display the accumulation of the flow of water around existing structures. The Bare Earth Model only displays the terrain of the ground surface, excluding objects on the ground such as trees or buildings, which would deviate the flow of the river flooding.

Floods can be monitored and analyzed without it being necessary to visit the site. This type of study demonstrates the vast implications of how floods can better be studied in the future. This methodology shows how the further evaluation and analysis of past floods better improve the understanding of how rising water levels affect the topography of a region like Schenectady. The ability to better understand how flooding affects neighborhoods like Stockade in Schenectady can result in improved measures that may be taken to prepare for and reduce the severity of the implications that result from flooding.

Conclusion

The integration of LiDAR data and Google Earth data into Global Mapper is able to construct a more accurate flood simulation, since this method combines the efficiency of Google's geolocation database along with the accuracy of LiDAR elevation data. While the utilization of the DTM is unique to GIS software such as Global Mapper, it is not necessary for our analysis since it does not display water flow around existing infrastructure; therefore, it cannot be an accurate model to determine flood damages for residents. This methodology is useful for future research because it provides a framework for how floods can be evaluated remotely, which can produce more prompt results on a broader scale.

References:

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Poster Presentation